

SuperBeam Project Summary

Project started 6 Apr 2019 MS

Site address: 95 Malvern Road

Job: Loft and alterations

Client: Damon

Job number: mal/sdc/01

ITEMS:

- 1: Beam: Rafters
Span: 1.5 m. Reactions: R1: 0.58 kN R2: 0.58 kN
Use 50 x 125 C16
C16 at 400 c/c

- 2: Beam: Flat roof joists
Span: 5.3 m. Reactions: R1: 1.86 kN R2: 1.86 kN
Use 50 x 220 C24
C24 at 400 c/c

- 3: Beam: Dormer Flat roof joists
Span: 2.0 m. Reactions: R1: 0.70 kN R2: 0.70 kN
Use 50 x 125 C24
C24 at 400 c/c

- 4: Beam: Rafters Back
Span: 3.5 m. Reactions: R1: 1.37 kN R2: 1.37 kN
Use 50 x 170 C16
C16 at 400 c/c

- 5: Beam: Dormer side rafters
Span: 2.5 m. Reactions: R1: 0.90 kN R2: 1.55 kN
Use 2no 50 x 125 C16

- 6: Beam: Dormer side rafter supports
Span: 2.0 m. Reactions: R1: 1.56 kN R2: 0.75 kN
Use 2no 50 x 125 C24

- 7: Beam: Ceiling collars
Span: 1.5 m. Reactions: R1: 0.15 kN R2: 0.15 kN
Use 47 x 125 C24
C24 at 400 c/c

- 8: Beam: Ridge beam 1
Span: 6.5 m. Reactions: R1: 12.07 kN R2: 11.96 kN
Use 152 x 152 x 44 UC S355
Bearing R1:
Column
Bearing R2: 15 mm m.s. bearing plate, size 300 x 100 mm

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- 9: Beam: Ridge beam 2
 Span: 6.5 m. Reactions: R1: 17.95 kN R2: 19.79 kN
 Use 203 x 203 x 46 UC S355
 Bearing R1:

Column

Bearing R2: 20 mm m.s. bearing plate, size 450 x 100 mm

- 10: Beam: Ridge beam 3
 Span: 7.6 m. Reactions: R1: 10.90 kN R2: 19.46 kN
 Use 203 x 203 x 46 UC S355
 Bearing R1:

Floor beam

Bearing R2: 20 mm m.s. bearing plate, size 450 x 100 mm

Beam cranks down pitch of hip

- 11: Beam: T1
 Span: 6.6 m. Reactions: R1: 45.53 kN R2: 11.09 kN
 Use 203 x 203 x 46 UC S355
 Bearing R1: 50 mm m.s. bearing plate, size 1050 x 100 mm
 Bearing R2: 10 mm m.s. bearing plate, size 250 x 100 mm

- 12: Beam: T2
 Span: 3.1 m. Reactions: R1: 20.31 kN R2: 15.52 kN
 Use 152 x 152 x 23 UC S355
 Bearing R1:

To floor beam

Bearing R2: 15 mm m.s. bearing plate, size 350 x 100 mm

- 13: Steel column: R1 of Ridge beam 2
 Unfactored load = 18.0 kN
 Moments due to load: $M_x = 2.87 \text{ kNm}$ $M_y = 0.00 \text{ kNm}$
 Section: 120 x 80 x 10 RHS Grade 50 on 200 x 205 x 10 mm base plate
 Sum of stress ratios = 0.160 OK
Weld plates to top and bottom of column. Bolt top plate to ridge beam via 4 No M16 bolts. Bottom plate 400x100x10 thick to beam via 4 No M16 bolts

- 14: Steel column: R1 of Ridge beam 1
 Unfactored load = 12.1 kN
 Moments due to load: $M_x = 1.93 \text{ kNm}$ $M_y = 0.00 \text{ kNm}$
 Section: 120 x 80 x 10 RHS Grade 50 on 200 x 205 x 10 mm base plate
 Sum of stress ratios = 0.108 OK
Weld plates to top and bottom of column. Bolt top plate to ridge beam via 4 No M16 bolts. Bottom plate 400x100x10 thick to beam via 4 No M16 bolts

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15: Beam: T3

Span: 2.0 m. Reactions: R1: 18.71 kN R2: 7.84 kN

Use 152 x 152 x 23 UC S355

Bearing R1: 20 mm m.s. bearing plate, size 450 x 100 mm

Bearing R2:

Web of T4 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

16: Beam: T4

Span: 4.0 m. Reactions: R1: 4.53 kN R2: 9.38 kN

Use 152 x 89 x 16 UB S355

Bearing R1:

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Bearing R2: 10 mm m.s. bearing plate, size 225 x 100 mm

17: Beam: T5

Span: 4.0 m. Reactions: R1: 4.11 kN R2: 4.87 kN

Use 152 x 89 x 16 UB S355

Bearing R1:

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Bearing R2: 10 mm m.s. bearing plate, size 225 x 100 mm

18: Beam: T6

Span: 6.0 m. Reactions: R1: 8.61 kN R2: 5.21 kN

Use 203 x 102 x 23 UB S355

Bearing R1:

Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Bearing R2:

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

19: Beam: T8

Span: 6.0 m. Reactions: R1: 7.93 kN R2: 10.53 kN

Use 203 x 133 x 25 UB S355

Bearing R1:

Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Bearing R2:

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

20: Beam: T7

Span: 6.0 m. Reactions: R1: 6.55 kN R2: 10.47 kN

Use 203 x 102 x 23 UB S355

Bearing R1:

Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Bearing R2:

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

21: Beam: FJ1

Span: 6.0 m. Reactions: R1: 1.79 kN R2: 2.27 kN

Use 75 x 220 C24

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22: Beam: FJ2
 Span: 6.0 m. Reactions: R1: 1.50 kN R2: 1.50 kN
 Use 50 x 220 C24

23: Beam: T9
 Span: 1.8 m. Reactions: R1: 2.43 kN R2: 2.43 kN
 Use 2no 47 x 220 C24

24: Beam: T10
 Span: 0.9 m. Reactions: R1: 1.48 kN R2: 1.48 kN
 Use 2no 47 x 220 C24
 Bearing R1:

Hanger to T5

Bearing R2:

Post to floor

25: Beam: T11
 Span: 2.0 m. Reactions: R1: 1.42 kN R2: 1.61 kN
 Use 2no 47 x 220 C24
 Bearing R1:
hanger to T11
 Bearing R2: 10 mm m.s. bearing plate, size 94 x 100 mm

26: Beam: T12
 Span: 2.0 m. Reactions: R1: 6.78 kN R2: 6.78 kN
 Use 2no 47 x 220 C24
 Bearing R1:
Hangere to T8
 Bearing R2: 10 mm m.s. bearing plate, size 175 x 100 mm

27: Beam: T13
 Span: 1.0 m. Reactions: R1: 1.64 kN R2: 1.64 kN
 Use 2no 50 x 220 C24

28: Beam: B2
 Span: 6.4 m. Reactions: R1: 38.46 kN R2: 33.63 kN
 Use 203 x 203 x 71 UC S355
 Bearing R1:
Web of T1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.
 Bearing R2: 30 mm m.s. bearing plate, size 650 x 120 mm

29: Beam: B1
 Span: 6.8 m. Reactions: R1: 70.95 kN R2: 37.72 kN
 Use 254 x 254 x 73 UC S355
 Bearing R1: 1600 x 100 mm padstone
Or shs
 Bearing R2: 40 mm m.s. bearing plate, size 850 x 100 mm

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30: Beam: I1
 Span: 1.1 m. Reactions: R1: 5.02 kN R2: 5.02 kN
 Use 203 x 203 x 46 UC S355
 Bearing R1: 10 mm m.s. bearing plate, size 204 x 205 mm
 Bearing R2: As R1

31: Beam: BA
 Span: 5.2 m. Reactions: R1: 6.47 kN R2: 6.47 kN
 Use 178 x 102 x 19 UB S355
 Bearing R1:

Web of Lower spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.
 Bearing R2: 10 mm m.s. bearing plate, size 150 x 100 mm

32: Beam: BB
 Span: 4.1 m. Reactions: R1: 5.10 kN R2: 5.10 kN
 Use 178 x 102 x 19 UB S355
 Bearing R1: 10 mm m.s. bearing plate, size 125 x 100 mm
 Bearing R2:

Web of Lower spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

33: Beam: Lower Ground Spine beam
 Span: 6.6 m. Reactions: R1: 56.21 kN R2: 59.80 kN
 Use 254 x 254 x 89 UC S355
 Bearing R1: 1050 x 120 mm padstone

Or SHS

Bearing R2: 1150 x 120 mm padstone

Or SHS

34: Beam: SHS1(R1 of lower ground floor beam)
 Span: 1.0 m. Reactions: R1: 28.33 kN R2: 28.33 kN
 Use 120 x 120 x 12.5 HF SHS S355 (40.9kg/m)

35: Beam: SHS2(R2 Of Lower ground floor beam)
 Span: 1.0 m. Reactions: R1: 30.13 kN R2: 30.13 kN
 Use 120 x 120 x 12.5 HF SHS S355 (40.9kg/m)

36: Beam: Upper Ground floor spine beam
 Span: 6.6 m. Reactions: R1: 77.83 kN R2: 89.94 kN
 Use 305 x 305 x 97 UC S355
 Bearing R1: 1750 x 100 x 75h mm padstone

Or SHS

Bearing R2: 2000 x 100 mm padstone

Or SHS

37: Beam: SHS3 (R1 of B1)
 Span: 1.0 m. Reactions: R1: 35.70 kN R2: 35.70 kN
 Use 120 x 120 x 12.5 HF SHS S355 (40.9kg/m)

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38: Beam: SHS4 (R2 of Upper ground floor beam)
Span: 1.0 m. Reactions: R1: 45.19 kN R2: 45.19 kN
Use 120 x 120 x 12.5 HF SHS S355 (40.9kg/m)

39: Beam: BC
Span: 6.5 m. Reactions: R1: 8.09 kN R2: 8.09 kN
Use 203 x 102 x 23 UB S355
Bearing R1:

Web of upper spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Bearing R2: 10 mm m.s. bearing plate, size 200 x 100 mm

40: Beam: BD
Span: 4.1 m. Reactions: R1: 5.10 kN R2: 5.10 kN
Use 127 x 76 x 13 UB S355
Bearing R1: 10 mm m.s. bearing plate, size 125 x 100 mm
Bearing R2:

Web of Upper spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

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Beam: Rafters**Span: 1.5 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U	T o.w.	0.1	0		L	0.08	0.08
U	T Pitched roof	1.7*.4	0		L	0.51	0.51
						0.58	0.58

Total load: 1.17 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 0.22 kNm at 0.75 m. from R1

Maximum S.F. = 0.58 kN at 0.00 m. from R1

Total deflection = $0.051 \times 10^8 / EI$ at 0.75 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C16 timber

Use 50 x 125 C16 2.3 kg/m approx $z = 130.2 \text{ cm}^3$ $I = 814 \text{ cm}^4$ Timber grade: C16 Load sharing system: $K_8 = 1.1$ [§2.10.11] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/125)^{0.11} = 1.10$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 8,800 \text{ N/mm}^2$ (E_{mean})**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 5.3 \times 1.00 \times 1.10 \times 1.1 = 6.42 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 0.219 \times 1000 / 130.2 = 1.68 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm, //} = t_{g, //} \cdot K_3 \cdot K_8 = 0.67 \times 1.00 \times 1.1 = 0.74 \text{ N/mm}^2$ Applied shear stress, $t_a = 0.585 \times 1000 \times 3 / (2 \times 50 \times 125) = 0.14 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.051 \times 10^8 / (8,800 \times 814) = 0.72 \text{ mm}$ Mid-span shear deflection = $1.2 \times 0.219 \times 10^6 / ((E/16) \times 50 \times 125) = 0.08 \text{ mm}$ Total deflection = $0.72 + 0.08 = 0.79 \text{ mm}$ ($0.0005 L$) $\leq 0.003L$ OK

C16 at 400 c/c

Beam: Flat roof joists**Span: 5.3 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.27	0.27
U T	Flat roof	1.5*.4	0		L	1.59	1.59
						1.86	1.86

Total load: 3.71 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 2.46 kNm at 2.65 m. from R1

Maximum S.F. = 1.85 kN at 0.00 m. from R1

Total deflection = $7.19 \times 10^8 / EI$ at 2.65 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 50 x 220 C24 4.6 kg/m approx $z = 403.3 \text{ cm}^3$ $I = 4,437 \text{ cm}^4$ Timber grade: C24 Load sharing system: $K_8 = 1.1$ [§2.10.11] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 10,800 \text{ N/mm}^2$ (E_{mean})**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 2.46 \times 1000 / 403 = 6.09 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 1.85 \times 1000 \times 3 / (2 \times 50 \times 220) = 0.25 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $7.19 \times 10^8 / (10,800 \times 4,437) = 15.01 \text{ mm}$ Mid-span shear deflection = $1.2 \times 2.46 \times 10^6 / ((E/16) \times 50 \times 220) = 0.40 \text{ mm}$ Total deflection = $15.01 + 0.40 = 15.40 \text{ mm}$ ($0.0029 L$) $\leq 0.003L$ OK

C24 at 400 c/c

Beam: Dormer Flat roof joists**Span: 2.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U	T o.w.	0.1	0		L	0.10	0.10
U	T Flat roof	1.5*.4	0		L	0.60	0.60
						0.70	0.70

Total load: 1.40 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 0.35 kNm at 1.00 m. from R1

Maximum S.F. = 0.70 kN at 0.00 m. from R1

Total deflection = $0.146 \times 10^8 / EI$ at 1.00 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 50 x 125 C24 2.6 kg/m approx $z = 130.2 \text{ cm}^3$ $I = 814 \text{ cm}^4$ Timber grade: C24 Load sharing system: $K_8 = 1.1$ [§2.10.11] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/125)^{0.11} = 1.10$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 10,800 \text{ N/mm}^2$ (E_{mean})**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.10 \times 1.1 = 9.08 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 0.350 \times 1000 / 130.2 = 2.69 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 0.700 \times 1000 \times 3 / (2 \times 50 \times 125) = 0.17 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.146 \times 10^8 / (10,800 \times 814) = 1.66 \text{ mm}$ Mid-span shear deflection = $1.2 \times 0.350 \times 10^6 / ((E/16) \times 50 \times 125) = 0.10 \text{ mm}$ Total deflection = $1.66 + 0.10 = 1.76 \text{ mm}$ (0.0009 L) $\leq 0.003L$ OK

C24 at 400 c/c

Beam: Rafters Back**Span: 3.5 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.17	0.17
U T	Pitched roof	1.7*.4	0		L	1.19	1.19
						<u>1.37</u>	<u>1.37</u>

Total load: 2.73 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 1.19 kNm at 1.75 m. from R1

Maximum S.F. = 1.36 kN at 0.00 m. from R1

Total deflection = $1.52 \times 10^8 / EI$ at 1.75 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C16 timber

Use 50 x 170 C16 3.1 kg/m approx $z = 240.8 \text{ cm}^3$ $I = 2,047 \text{ cm}^4$ Timber grade: C16 Load sharing system: $K_8 = 1.1$ [§2.10.11] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/170)^{0.11} = 1.06$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 8,800 \text{ N/mm}^2$ (E_{mean})**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 5.3 \times 1.00 \times 1.06 \times 1.1 = 6.21 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 1.19 \times 1000 / 240.8 = 4.96 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.67 \times 1.00 \times 1.1 = 0.74 \text{ N/mm}^2$ Applied shear stress, $t_a = 1.36 \times 1000 \times 3 / (2 \times 50 \times 170) = 0.24 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $1.52 \times 10^8 / (8,800 \times 2,047) = 8.46 \text{ mm}$ Mid-span shear deflection = $1.2 \times 1.19 \times 10^6 / ((E/16) \times 50 \times 170) = 0.31 \text{ mm}$ Total deflection = $8.46 + 0.31 = 8.77 \text{ mm}$ ($0.0025 L$) $\leq 0.003L$ OK

C16 at 400 c/c

Beam: Dormer side rafters**Span: 2.5 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U	T o.w.	0.1	0		L	0.13	0.13
V	T Dormer cheek	1.1*.1	0	1.1*1.5	2.5	0.78	1.42
						0.90	1.55

Total load: 2.45 kN

Load types: U: UDL V: Variable load T: Total (positions in m. from R1)

Maximum B.M. = 0.78 kNm at 1.41 m. from R1

Maximum S.F. = -1.55 kN at R2

Total deflection = $0.499 \times 10^8 / EI$ at 1.29 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C16 timber

Use 2no 50 x 125 C16 4.6 kg/m approx

$z = 260.4 \text{ cm}^3$ $I = 1,628 \text{ cm}^4$

Timber grade: C16 2 members acting together: $K_8 = 1.1$ [§2.9]

K_3 (loading duration factor) = 1.00 (long term)

K_7 (depth factor) = $(300/125)^{0.11} = 1.10$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10]

$E = 5,800 \times 1.14 = 6,612 \text{ N/mm}^2$ ($E_{min} \cdot K_9$)

Bending

Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 5.3 \times 1.00 \times 1.10 \times 1.1 = 6.42 \text{ N/mm}^2$

Applied bending stress, $s_{m,a} = 0.778 \times 1000/260 = 2.99 \text{ N/mm}^2$ OK

Shear

Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.67 \times 1.00 \times 1.1 = 0.74 \text{ N/mm}^2$

Applied shear stress, $t_a = 1.55 \times 1000 \times 3 / (2 \times 100 \times 125) = 0.19 \text{ N/mm}^2$ OK

Deflection

Bending deflection = $0.499 \times 10^8 / (6,612 \times 1,628) = 4.64 \text{ mm}$

Mid-span shear deflection = $1.2 \times 0.766 \times 10^6 / ((E/16) \times 100 \times 125) = 0.18 \text{ mm}$

Total deflection = $4.64 + 0.18 = 4.81 \text{ mm}$ ($0.0019 L$) $\leq 0.003L$ OK

Beam: Dormer side rafter supports**Span: 2.0 m.**

Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T o.w.	0.1	0		L	0.10	0.10
U T Flat roof	1.5*.4	0		L	0.60	0.60
P T Beam: Dormer side : R1	0.90 [B/F]	.1			0.86	0.05
					1.56	0.75

Total load: 2.30 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 0.40 kNm at 0.94 m. from R1

Maximum S.F. = 1.56 kN at 0.00 m. from R1

Total deflection = $0.168 \times 10^8 / EI$ at 0.98 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 2no 50 x 125 C24 5.3 kg/m approx $z = 260.4 \text{ cm}^3$ $I = 1,628 \text{ cm}^4$ Timber grade: C24 2 members acting together: $K_8 = 1.1$ [§2.9] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/125)^{0.11} = 1.10$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 7,200 \times 1.14 = 8,208 \text{ N/mm}^2$ ($E_{min} \cdot K_9$)**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.10 \times 1.1 = 9.08 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 0.397 \times 1000/260 = 1.52 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,II} = t_{g,II} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 1.56 \times 1000 \times 3/(2 \times 100 \times 125) = 0.19 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.168 \times 10^8 / (8,208 \times 1,628) = 1.26 \text{ mm}$ Mid-span shear deflection = $1.2 \times 0.395 \times 10^6 / ((E/16) \times 100 \times 125) = 0.07 \text{ mm}$ Total deflection = $1.26 + 0.07 = 1.33 \text{ mm}$ ($0.0007 L$) $\leq 0.003L$ OK

Beam: Ceiling collars**Span: 1.5 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.08	0.08
U T	Ceiling	.25*.4	0		L	0.08	0.08
						0.15	0.15

Total load: 0.30 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 0.06 kNm at 0.75 m. from R1

Maximum S.F. = 0.15 kN at 0.00 m. from R1

Total deflection = $0.013 \times 10^8/EI$ at 0.75 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 47 x 125 C24 2.5 kg/m approx $z = 122.4 \text{ cm}^3$ $I = 765 \text{ cm}^4$ Timber grade: C24 Load sharing system: $K_8 = 1.1$ [§2.10.11] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/125)^{0.11} = 1.10$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 10,800 \text{ N/mm}^2$ (E_{mean})**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.10 \times 1.1 = 9.08 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 0.056 \times 1000/122.4 = 0.46 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 0.150 \times 1000 \times 3/(2 \times 47 \times 125) = 0.04 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.013 \times 10^8/(10,800 \times 765) = 0.16 \text{ mm}$ Mid-span shear deflection = $1.2 \times 0.056 \times 10^6/((E/16) \times 47 \times 125) = 0.02 \text{ mm}$ Total deflection = $0.16 + 0.02 = 0.18 \text{ mm}$ ($0.0001 L$) $\leq 0.003L$ OK

C24 at 400 c/c

Beam: Ridge beam 1**Span: 6.5 m.**

Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T o.w.	0.45	0		L	1.46	1.46
U T Apex	1.7*1/2	0		L	2.76	2.76
R T Pitched roof	1.7*2.5/2	0		1.5	2.82	0.37
R T Flat roof	1.5*2/2	1.5		4.6	2.47	2.18
R T Pitched roof	1.7*2.5/2	4.6		6.5	0.59	3.45
R T Ceiling collars	.25*1.5/2	1.5		4.6	0.31	0.27
P T Beam: Dormer side : R1	1.56 [B/F]	1.5			1.20	0.36
P T Beam: Dormer side : R1	1.56 [B/F]	4.6			0.46	1.10
					12.07	11.96

Total load: 24.02 kN

Load types: U: UDL R: Part UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 19.07 kNm at 3.30 m. from R1

Maximum S.F. = 12.07 kN at 0.00 m. from R1

Total deflection = $85.3 \times 10^8 / EI$ at 3.25 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 152 x 152 x 44 UC S355

$D=166.0$ mm $B=155.9$ mm $t=9.5$ mm $T=13.6$ mm $I_x=2,703$ cm⁴ $r_y=3.92$ cm $Z_x=326$ cm³

Bending

$L_E/r_y = 6.50 \times 100/3.92 = 166$ $D/T = 12.2$

Permissible bending stress, $p_{bc} = 146.4$ N/mm² (Table 3b)

Actual bending stress, $f_{bc} = 19.07 \times 1000/326 = 58.5$ N/mm² OK

Shear

Maximum shear in web, $f_s = 12.07 \times 1000/(9.5 \times 166.0) = 7.7$ N/mm² OK

Beam web

Check unstiffened web capacities with loads of 12.07 kN and 11.96 kN

Bearing: $p_b = 260$ N/mm² (Table 9); $C1 = 90.7$ kN; $C2 = 2.47$ kN/mm

Buckling: $p_c = 203$ N/mm² (Table 17b); $C1 = 160$ kN; $C2 = 1.93$ kN/mm

R1: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 90.7$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 160$ kN

R2: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 90.7$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 160$ kN

Deflection

Total deflection = $85.3 \times 1e8/(205,000 \times 2,703) = 15.4$ mm ($L/422$) OK

Combined bending and shear check (14.c)

Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.160$ at 3.25 m. (≤ 1.25 OK) [14.c]

Bearings (bearing plate sizing to BS5950-1:2000)

152 x 152 x 44 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 48.9 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-entered

Factored reaction = $12.07 \times 1.55 = 18.70 \text{ kN}$

Column

R2: 300 x 100 mm bearing plate

Factored reaction = $11.96 \times 1.55 = 18.53 \text{ kN}$

15 mm m.s. bearing plate, size 300 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(300 - 48.9)/2 = 125.6 \text{ mm}$

Factored stress under plate = $1.55 \times 11.96 \times 1000 / (300 \times 100) = 0.62 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.62 \times 126 \times 126 / 275)} = 10.3 \text{ mm}$: use 15mm

Factored bending stress in plate = $0.62 \times 126 \times (126/2) / (15 \times 15/6) = 129.9 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: Ridge beam 2**Span: 6.5 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.5	0		L	1.63	1.63
U T	Apex	1.7*1	0		L	5.53	5.53
R T	Pitched roof	1.7*2.8/2	0		1.5	3.16	0.41
R T	Flat roof	1.5*5.3/2	1.5		6.5	7.64	12.23
						17.95	19.79

Total load: 37.75 kN

Load types: U: UDL R: Part UDL T: Total (positions in m. from R1)

Maximum B.M. = 31.72 kNm at 3.29 m. from R1

Maximum S.F. = -19.79 kN at R2

Total deflection = $139 \times 10^8 / EI$ at 3.25 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 203 x 46 UC S355D=203.2 mm B=203.6 mm t=7.2 mm T=11.0 mm $I_x=4,570 \text{ cm}^4$ $r_y=5.13 \text{ cm}$ $Z_x=450 \text{ cm}^3$ **Bending** $L_E/r_y = 6.50 \times 100/5.13 = 127$ D/T = 18.5Permissible bending stress, $p_{bc} = 143.9 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 31.72 \times 1000/450 = 70.5 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 19.79 \times 1000/(7.2 \times 203.2) = 13.5 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 17.95 kN and 19.79 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 68.7 \text{ kN}$; $C2 = 1.87 \text{ kN/mm}$ Buckling: $p_c = 194 \text{ N/mm}^2$ (Table 17b); $C1 = 142 \text{ kN}$; $C2 = 1.40 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 68.7 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 142 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 68.7 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 142 \text{ kN}$ **Deflection**Total deflection = $139 \times 1e8/(205,000 \times 4,570) = 14.8 \text{ mm}$ (L/438) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.240$ at 3.25 m. (≤ 1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)203 x 203 x 46 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 45.5 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-enteredFactored reaction = $17.95 \times 1.55 = 27.83 \text{ kN}$

Column

R2: 450 x 100 mm bearing plate

Factored reaction = $19.79 \times 1.55 = 30.68 \text{ kN}$

20 mm m.s. bearing plate, size 450 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(450 - 45.5)/2 = 202.2 \text{ mm}$

Factored stress under plate = $1.55 \times 19.79 \times 1000 / (450 \times 100) = 0.68 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.68 \times 202 \times 202 / 265)} = 17.8 \text{ mm}$: use 20mm

Factored bending stress in plate = $0.68 \times 202 \times (202/2) / (20 \times 20/6) = 209.1 \text{ N/mm}^2$ ($p_y = 265 \text{ N/mm}^2$)

Beam: Ridge beam 3**Span: 7.6 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.5	0		L	1.90	1.90
R T	Pitched roof	1.7*.4	.8		2.8	1.04	0.32
R T	Pitched roof	1.7*1.5/2	2.8		7.6	1.93	4.19
R T	Flat roof	1.5*5.3/2	2.8		7.6	6.03	13.05
						10.90	19.46

Total load: 30.36 kN

Load types: U: UDL R: Part UDL T: Total (positions in m. from R1)

Maximum B.M. = 32.94 kNm at 4.21 m. from R1

Maximum S.F. = -19.46 kN at R2

Total deflection = $190 \times 10^8 / EI$ at 3.93 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 203 x 46 UC S355D=203.2 mm B=203.6 mm t=7.2 mm T=11.0 mm $I_x=4,570 \text{ cm}^4$ $r_y=5.13 \text{ cm}$ $Z_x=450 \text{ cm}^3$ **Bending** $L_E/r_y = 7.60 \times 100/5.13 = 148$ D/T = 18.5Permissible bending stress, $p_{bc} = 125.3 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 32.94 \times 1000/450 = 73.2 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 19.46 \times 1000/(7.2 \times 203.2) = 13.3 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 10.90 kN and 19.46 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 68.7 \text{ kN}$; $C2 = 1.87 \text{ kN/mm}$ Buckling: $p_c = 194 \text{ N/mm}^2$ (Table 17b); $C1 = 142 \text{ kN}$; $C2 = 1.40 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 68.7 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 142 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 68.7 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 142 \text{ kN}$ **Deflection**Total deflection = $190 \times 1e8/(205,000 \times 4,570) = 20.3 \text{ mm}$ ($L/374$) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.341$ at 4.26 m. (≤ 1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)203 x 203 x 46 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 45.5 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-enteredFactored reaction = $10.90 \times 1.55 = 16.89 \text{ kN}$

Floor beam

R2: 450 x 100 mm bearing plate

Factored reaction = $19.46 \times 1.55 = 30.17 \text{ kN}$

20 mm m.s. bearing plate, size 450 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(450 - 45.5)/2 = 202.2 \text{ mm}$

Factored stress under plate = $1.55 \times 19.46 \times 1000 / (450 \times 100) = 0.67 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.67 \times 202 \times 202 / 265)} = 17.6 \text{ mm}$: use 20mm

Factored bending stress in plate = $0.67 \times 202 \times (202/2) / (20 \times 20/6) = 205.7 \text{ N/mm}^2$ ($p_y = 265 \text{ N/mm}^2$)

Beam cranks down pitch of hip

Beam: T1**Span: 6.6 m.**

Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T o.w.	0.5	0		L	1.65	1.65
U T Floor	1.5*.4	0		L	1.98	1.98
P T Beam: Ridge beam 3 : R1	10.90 [B/F]	2.4			6.93	3.96
P T B2	38.46	.6			34.96	3.50
					45.53	11.09

Total load: 56.62 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 36.87 kNm at 2.40 m. from R1

Maximum S.F. = 45.53 kN at 0.00 m. from R1

Total deflection = $149 \times 10^8 / EI$ at 3.02 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 203 x 46 UC S355D=203.2 mm B=203.6 mm t=7.2 mm T=11.0 mm $I_x=4,570 \text{ cm}^4$ $r_y=5.13 \text{ cm}$ $Z_x=450 \text{ cm}^3$ **Bending** $L_E/r_y = 6.60 \times 100/5.13 = 129$ D/T = 18.5Permissible bending stress, $p_{bc} = 142.1 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 36.87 \times 1000/450 = 81.9 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 45.53 \times 1000/(7.2 \times 203.2) = 31.1 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 45.53 kN and 11.09 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 68.7 \text{ kN}$; $C2 = 1.87 \text{ kN/mm}$ Buckling: $p_c = 194 \text{ N/mm}^2$ (Table 17b); $C1 = 142 \text{ kN}$; $C2 = 1.40 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 68.7 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 142 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 68.7 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 142 \text{ kN}$ **Deflection**Total deflection = $149 \times 1e8/(205,000 \times 4,570) = 15.9 \text{ mm}$ (L/414) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.331$ at 2.38 m. (≤ 1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)203 x 203 x 46 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 45.5 \text{ mm}$

Factor reactions by 1.55 (user selected value)

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

R1: 1050 x 100 mm bearing plate

Factored reaction = $45.53 \times 1.55 = 70.57 \text{ kN}$

50 mm m.s. bearing plate, size 1050 x 100 mm

Bearing plate projection beyond stiff bearing length = $(1050 - 45.5)/2 = 502.2 \text{ mm}$

Factored stress under plate = $1.55 \times 45.53 \times 1000 / (1050 \times 100) = 0.67 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.67 \times 502 \times 502 / 255)} = 44.7 \text{ mm}$: use 50 mm

Factored bending stress in plate = $0.67 \times 502 \times (502/2) / (50 \times 50/6) = 203.4 \text{ N/mm}^2$ ($p_y = 255 \text{ N/mm}^2$)

R2: 250 x 100 mm bearing plate

Factored reaction = $11.09 \times 1.55 = 17.19 \text{ kN}$

10 mm m.s. bearing plate, size 250 x 100 mm

Bearing plate projection beyond stiff bearing length = $(250 - 45.5)/2 = 102.2 \text{ mm}$

Factored stress under plate = $1.55 \times 11.09 \times 1000 / (250 \times 100) = 0.69 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.69 \times 102 \times 102 / 275)} = 8.85 \text{ mm}$: use 10 mm

Factored bending stress in plate = $0.69 \times 102 \times (102/2) / (10 \times 10/6) = 215.6 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: T2**Span: 3.1 m.**

Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T o.w.	0.4	0		L	0.62	0.62
U T Floor	1.5*.4	0		L	0.93	0.93
U T Partition	.35*2.5	0		L	1.36	1.36
P T Beam: Ridge beam 2 : R1	17.95 [B/F]	.7			13.90	4.05
P T Beam: Ridge beam 1 : R1	12.07 [B/F]	2.2			3.50	8.56
					20.31	15.52

Total load: 35.83 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 14.05 kNm at 1.26 m. from R1

Maximum S.F. = 20.31 kN at 0.00 m. from R1

Total deflection = $15.1 \times 10^{-8}/EI$ at 1.52 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 152 x 152 x 23 UC S355D=152.4 mm B=152.2 mm t=5.8 mm T=6.8 mm $I_x=1,250 \text{ cm}^4$ $r_y=3.70 \text{ cm}$ $Z_x=164 \text{ cm}^3$ **Bending** $L_E/r_y = 3.10 \times 100/3.70 = 84$ D/T = 22.4Permissible bending stress, $p_{bc} = 182.2 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 14.05 \times 1000/164 = 85.6 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 20.31 \times 1000/(5.8 \times 152.4) = 23.0 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 20.31 kN and 15.52 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 37.6 \text{ kN}$; $C2 = 1.51 \text{ kN/mm}$ Buckling: $p_c = 196 \text{ N/mm}^2$ (Table 17b); $C1 = 86.6 \text{ kN}$; $C2 = 1.14 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 37.6 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 86.6 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 37.6 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 86.6 \text{ kN}$ **Deflection**Total deflection = $15.1 \times 10^{-8}/(205,000 \times 1,250) = 5.9 \text{ mm}$ ($L/527$) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.225$ at 0.68 m. ($<=1.25$ OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)152 x 152 x 23 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 31.6 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-enteredFactored reaction = $20.31 \times 1.55 = 31.48 \text{ kN}$

To floor beam

R2: 350 x 100 mm bearing plate

Factored reaction = $15.52 \times 1.55 = 24.06 \text{ kN}$

15 mm m.s. bearing plate, size 350 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(350 - 31.6)/2 = 159.2 \text{ mm}$

Factored stress under plate = $1.55 \times 15.52 \times 1000 / (350 \times 100) = 0.69 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.69 \times 159 \times 159 / 275)} = 13.8 \text{ mm}$: use 15mm

Factored bending stress in plate = $0.69 \times 159 \times (159/2) / (15 \times 15/6) = 232.4 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

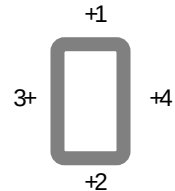
Steel column calculation to BS449 Part 2 using S355 steel

Location: R1 of Ridge beam 2

Length: 2.3 m.

Pos	Load	Total kN	Offset mm	M_{xx}	M_{yy}
2	Beam: Ridge beam 2 : R1	[B/F] 17.95	100	-2.87	0.00
	Total load	17.95		2.87	0.00

Load offsets are measured in mm. from faces of member; moments in kNm



SECTION SIZE : 120 x 80 x 10 HF RHS S355 (27.4kg/m)

Gross area, $A_g = 34.9 \text{ cm}^2$ $B = 80.0 \text{ mm}$ $D = 120.0 \text{ mm}$ $T = 10.0 \text{ mm}$ $t = 10.0 \text{ mm}$
 $r_x = 4.18 \text{ cm}$ $r_y = 2.99 \text{ cm}$ $Z_x = 102 \text{ cm}^3$ $Z_y = 78.3 \text{ cm}^3$

Compression

Major axis: $L_{Ex} = 1.0L = 2.30 \text{ m}$. Slenderness, $\lambda_x = 2.30 \times 100/4.18 = 55.0$

Minor axis: $L_{Ey} = 1.0L = 2.30 \text{ m}$. Slenderness, $\lambda_y = 2.30 \times 100/2.99 = 76.9$

Permissible compressive stress, $p_c = 135.2 \text{ N/mm}^2$ [Table 17b]

Actual compressive stress, $F_c = 17.95 \times 1000/(34.9 \times 100) = 5.14 \text{ N/mm}^2$

Compressive resistance = $A_g \cdot p_c = 34.9 \times 100 \times 135.2/1000 = 472 \text{ kN}$

Bending about major axis

Permissible bending stress, $p_{bc,x} = 230.0 \text{ N/mm}^2$ [19.d, Table 2]

Actual bending stress, $F_{bc,x} = 2.87 \times 1000/102.0 = 28.16 \text{ N/mm}^2$

Bending about minor axis

Permissible bending stress, $p_{bc,y} = 230.0 \text{ N/mm}^2$ [19.d, Table 2]

Actual bending stress, $F_{bc,y} = 0.00 \times 1000/78.3 = 0.000 \text{ N/mm}^2$

Summary

$$F/A_g \cdot p_c = 17.95/472 = 0.038 \text{ [1]}$$

$$F_{bc,x}/P_{bc,x} = 28.2/230.0 = 0.122 \text{ [2]}$$

$$F_{bc,y}/P_{bc,y} = 0.00/230.0 = 0.000 \text{ [3]}$$

Sum of stress ratios [1] + [2] + [3] = **0.160** OK [14.a]

Baseplate calculation (considering axial load only)

Factor load by 1.40 (user selected value): factored load on base = 25.13 kN

Concrete strength, $f_{cu} = 20 \text{ N/mm}^2$

Minimum area required = $F_c/0.6f_{cu} = 25.13 \times 1000/(0.6 \times 20) = 2,094 \text{ mm}^2$

K (min reqd proj) = -2.09mm Minimum base plate size = 120 x 80mm

Minimum thickness = $K\sqrt{3 \times 0.6 \cdot f_{cu}/p_{yp}} = -0.757 \text{ mm}$ ($p_{yp} = 275 \text{ N/mm}^2$)

Use 200 x 205 x 10mm base plate

Weld plates to top and bottom of column. Bolt top plate to ridge beam via 4 No M16 bolts. Bottom plate 400x100x10thick to beam via 4 No M16 bolts

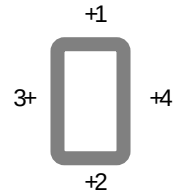
Steel column calculation to BS449 Part 2 using S355 steel

Location: R1 of Ridge beam1

Length: 2.3 m.

Pos	Load	Total kN	Offset mm	M_{xx}	M_{yy}
2	Beam: Ridge beam 1 : R1	[B/F] 12.07	100	-1.93	0.00
	Total load	12.07		1.93	0.00

Load offsets are measured in mm. from faces of member; moments in kNm



SECTION SIZE : 120 x 80 x 10 HF RHS S355 (27.4kg/m)

Gross area, $A_g = 34.9 \text{ cm}^2$ $B = 80.0 \text{ mm}$ $D = 120.0 \text{ mm}$ $T = 10.0 \text{ mm}$ $t = 10.0 \text{ mm}$
 $r_x = 4.18 \text{ cm}$ $r_y = 2.99 \text{ cm}$ $Z_x = 102 \text{ cm}^3$ $Z_y = 78.3 \text{ cm}^3$

Compression

Major axis: $L_{Ex} = 1.0L = 2.30 \text{ m}$. Slenderness, $\lambda_x = 2.30 \times 100/4.18 = 55.0$

Minor axis: $L_{Ey} = 1.0L = 2.30 \text{ m}$. Slenderness, $\lambda_y = 2.30 \times 100/2.99 = 76.9$

Permissible compressive stress, $p_c = 135.2 \text{ N/mm}^2$ [Table 17b]

Actual compressive stress, $F_c = 12.07 \times 1000/(34.9 \times 100) = 3.46 \text{ N/mm}^2$

Compressive resistance = $A_g \cdot p_c = 34.9 \times 100 \times 135.2/1000 = 472 \text{ kN}$

Bending about major axis

Permissible bending stress, $p_{bc,x} = 230.0 \text{ N/mm}^2$ [19.d, Table 2]

Actual bending stress, $F_{bc,x} = 1.93 \times 1000/102.0 = 18.93 \text{ N/mm}^2$

Bending about minor axis

Permissible bending stress, $p_{bc,y} = 230.0 \text{ N/mm}^2$ [19.d, Table 2]

Actual bending stress, $F_{bc,y} = 0.00 \times 1000/78.3 = 0.000 \text{ N/mm}^2$

Summary

$$F/A_g \cdot p_c = 12.07/472 = 0.026 \text{ [1]}$$

$$F_{bc,x}/P_{bc,x} = 18.9/230.0 = 0.082 \text{ [2]}$$

$$F_{bc,y}/P_{bc,y} = 0.00/230.0 = 0.000 \text{ [3]}$$

Sum of stress ratios [1] + [2] + [3] = **0.108** OK [14.a]

Baseplate calculation (considering axial load only)

Factor load by 1.40 (user selected value): factored load on base = 16.89 kN

Concrete strength, $f_{cu} = 20 \text{ N/mm}^2$

Minimum area required = $F_c/0.6f_{cu} = 16.89 \times 1000/(0.6 \times 20) = 1,408 \text{ mm}^2$

K (min reqd proj) = -3.04mm Minimum base plate size = 120 x 80mm

Minimum thickness = $K\sqrt{3 \times 0.6 \cdot f_{cu}/p_{yp}} = -1.10 \text{ mm}$ ($p_{yp} = 275 \text{ N/mm}^2$)

Use 200 x 205 x 10mm base plate

Weld plates to top and bottom of column. Bolt top plate to ridge beam via 4 No M16 bolts. Bottom plate 400x100x10thick to beam via 4 No M16 bolts

Beam: T3**Span: 2.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.25	0.25
U T	Partition	.35*1.7	0		L	0.60	0.60
U T	Pitched roof	1.7/1/2	0		L	0.85	0.85
U T	Eaves	1.7*1.2/2	0		L	1.02	1.02
U T	Floor	2*2.8/2	0		L	2.80	2.80
P T	Beam: T2 : R2	15.52 [B/F]	.3			13.19	2.33
						18.71	7.84

Total load: 26.55 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 5.58 kNm at 0.58 m. from R1

Maximum S.F. = 18.71 kN at 0.00 m. from R1

Total deflection = $2.29 \times 10^{-8}/EI$ at 0.93 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 152 x 152 x 23 UC S355D=152.4 mm B=152.2 mm t=5.8 mm T=6.8 mm $I_x=1,250 \text{ cm}^4$ $r_y=3.70 \text{ cm}$ $Z_x=164 \text{ cm}^3$ **Bending** $L_E/r_y = 2.00 \times 100/3.70 = 54$ D/T = 22.4Permissible bending stress, $p_{bc} = 225.2 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 5.58 \times 1000/164 = 34.0 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 18.71 \times 1000/(5.8 \times 152.4) = 21.2 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 18.71 kN and 7.843 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 37.6 \text{ kN}$; $C2 = 1.51 \text{ kN/mm}$ Buckling: $p_c = 196 \text{ N/mm}^2$ (Table 17b); $C1 = 86.6 \text{ kN}$; $C2 = 1.14 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 37.6 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 86.6 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 37.6 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 86.6 \text{ kN}$ **Deflection**Total deflection = $2.29 \times 10^{-8}/(205,000 \times 1,250) = 0.9 \text{ mm}$ (L/2238) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.038$ at 0.28 m. (≤ 1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)152 x 152 x 23 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 31.6 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: 450 x 100 mm bearing plate

Factored reaction = $18.71 \times 1.55 = 29.00 \text{ kN}$

20 mm m.s. bearing plate, size 450 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(450 - 31.6)/2 = 209.2 \text{ mm}$

Factored stress under plate = $1.55 \times 18.71 \times 1000 / (450 \times 100) = 0.64 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.64 \times 209 \times 209 / 265)} = 17.9 \text{ mm}$: use 20mm

Factored bending stress in plate = $0.64 \times 209 \times (209/2) / (20 \times 20/6) = 211.6 \text{ N/mm}^2$ ($p_y = 265 \text{ N/mm}^2$)

R2: User-entered

Factored reaction = $7.84 \times 1.55 = 12.16 \text{ kN}$

Web of T4 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Beam: T4**Span: 4.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.50	0.50
U T	Floor	2*.4	0		L	1.60	1.60
V T	Dormer cheek	1.1*1.9	2.9	1.1*1.2	4	0.28	1.60
P T	Beam: T3 : R2	7.84 [B/F]	2.9			2.16	5.69
						<u>4.53</u>	<u>9.38</u>

Total load: 13.92 kN

Load types: U: UDL V: Variable load P: PL T: Total (positions in m. from R1)

Maximum B.M. = 8.73 kNm at 2.90 m. from R1

Maximum S.F. = -9.38 kN at R2

Total deflection = $12.4 \times 10^{-8}/EI$ at 2.17 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 152 x 89 x 16 UB S355

$D=152.4$ mm $B=88.7$ mm $t=4.5$ mm $T=7.7$ mm $I_x=834$ cm^4 $r_y=2.10$ cm $Z_x=109$ cm^3

Bending

$L_E/r_y = 4.00 \times 100/2.10 = 190$ $D/T = 19.8$

Permissible bending stress, $p_{bc} = 96.5$ N/mm^2 (Table 3b)

Actual bending stress, $f_{bc} = 8.73 \times 1000/109 = 80.1$ N/mm^2 OK

Shear

Maximum shear in web, $f_s = 9.38 \times 1000/(4.5 \times 152.4) = 13.7$ N/mm^2 OK

Beam web

Check unstiffened web capacities with loads of 4.534 kN and 9.385 kN

Bearing: $p_b = 260$ N/mm^2 (Table 9); $C1 = 31.0$ kN; $C2 = 1.17$ kN/mm

Buckling: $p_c = 187$ N/mm^2 (Table 17b); $C1 = 64.2$ kN; $C2 = 0.842$ kN/mm

R1: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 31.0$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 64.2$ kN

R2: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 31.0$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 64.2$ kN

Deflection

Total deflection = $12.4 \times 1e8/(205,000 \times 834) = 7.3$ mm (L/551) OK

Combined bending and shear check (14.c)

Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.685$ at 2.88 m. (≤ 1.25 OK) [14.c]

Bearings (bearing plate sizing to BS5950-1:2000)

152 x 89 x 16 UB stiff bearing length, $b_1 = t + 1.6r + 2T = 32.1$ mm

Factor reactions by 1.55 (user selected value)

R1: User-entered

Factored reaction = $4.53 \times 1.55 = 7.03$ kN

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: 225 x 100 mm bearing plate

Factored reaction = $9.38 \times 1.55 = 14.55 \text{ kN}$

10 mm m.s. bearing plate, size 225 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(225 - 32.1)/2 = 96.5 \text{ mm}$

Factored stress under plate = $1.55 \times 9.38 \times 1000 / (225 \times 100) = 0.65 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.65 \times 96.5 \times 96.5 / 275)} = 8.10 \text{ mm}$: use 10mm

Factored bending stress in plate = $0.65 \times 96.5 \times (96.5/2) / (10 \times 10/6) = 180.5 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: T5**Span: 4.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.50	0.50
U T	Floor	2*.4	0		L	1.60	1.60
V T	Dormer cheek	1.1*1.9	2.9	1.1*1.2	4	0.28	1.60
P T	T11	1.42	.8			1.14	0.28
P T	T10	1.48	2.4			0.59	0.89
						4.11	4.87

Total load: 8.98 kN

Load types: U: UDL V: Variable load P: PL T: Total (positions in m. from R1)

Maximum B.M. = 4.56 kNm at 2.40 m. from R1

Maximum S.F. = -4.87 kN at R2

Total deflection = $7.49 \times 10^{-8}/EI$ at 2.03 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 152 x 89 x 16 UB S355D=152.4 mm B=88.7 mm t=4.5 mm T=7.7 mm $I_x=834 \text{ cm}^4$ $r_y=2.10 \text{ cm}$ $Z_x=109 \text{ cm}^3$ **Bending** $L_E/r_y = 4.00 \times 100/2.10 = 190$ D/T = 19.8Permissible bending stress, $p_{bc} = 96.5 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 4.56 \times 1000/109 = 41.8 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 4.87 \times 1000/(4.5 \times 152.4) = 7.1 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 4.105 kN and 4.870 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 31.0 \text{ kN}$; $C2 = 1.17 \text{ kN/mm}$ Buckling: $p_c = 187 \text{ N/mm}^2$ (Table 17b); $C1 = 64.2 \text{ kN}$; $C2 = 0.842 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 31.0 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 64.2 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 31.0 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 64.2 \text{ kN}$ **Deflection**Total deflection = $7.49 \times 10^{-8}/(205,000 \times 834) = 4.4 \text{ mm}$ (L/913) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.188$ at 2.40 m. (<1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)152 x 89 x 16 UB stiff bearing length, $b_1 = t + 1.6r + 2T = 32.1 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-enteredFactored reaction = $4.11 \times 1.55 = 6.36 \text{ kN}$

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: 225 x 100 mm bearing plate

Factored reaction = $4.87 \times 1.55 = 7.55 \text{ kN}$

10 mm m.s. bearing plate, size 225 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(225 - 32.1)/2 = 96.5 \text{ mm}$

Factored stress under plate = $1.55 \times 4.87 \times 1000 / (225 \times 100) = 0.34 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.34 \times 96.5 \times 96.5 / 275)} = 5.84 \text{ mm}$: use 10mm

Factored bending stress in plate = $0.34 \times 96.5 \times (96.5/2) / (10 \times 10/6) = 93.7 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: T6**Span: 6.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.75	0.75
U T	Floor	2*.4	0		L	2.40	2.40
V T	Partition	1.1*1.4	0	1.1*2.5	1.2	2.29	0.28
V T	Partition	1.1*2.5	1.2	1.1*1.5	2.7	2.26	1.04
P T	T13	1.64	2.7			0.90	0.74
						8.61	5.21

Total load: 13.81 kN

Load types: U: UDL V: Variable load P: PL T: Total (positions in m. from R1)

Maximum B.M. = 11.47 kNm at 2.66 m. from R1

Maximum S.F. = 8.61 kN at 0.00 m. from R1

Total deflection = $40.6 \times 10^{-8}/EI$ at 2.85 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 102 x 23 UB S355D=203.2 mm B=101.8 mm t=5.4 mm T=9.3 mm $I_x=2,110 \text{ cm}^4$ $r_y=2.36 \text{ cm}$ $Z_x=207 \text{ cm}^3$ **Bending** $L_E/r_y = 6.00 \times 100/2.36 = 254$ D/T = 21.8Permissible bending stress, $p_{bc} = 70.1 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 11.47 \times 1000/207 = 55.4 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 8.61 \times 1000/(5.4 \times 203.2) = 7.8 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 8.606 kN and 5.208 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 41.1 \text{ kN}$; $C2 = 1.40 \text{ kN/mm}$ Buckling: $p_c = 178 \text{ N/mm}^2$ (Table 17b); $C1 = 97.8 \text{ kN}$; $C2 = 0.963 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 41.1 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 97.8 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 41.1 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 97.8 \text{ kN}$ **Deflection**Total deflection = $40.6 \times 10^{-8}/(205,000 \times 2,110) = 9.4 \text{ mm}$ (L/638) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.625$ at 2.64 m. (<1.25 OK) [14.c]**Bearings****R1: User-entered**Factored reaction = $8.61 \times 1.55 = 13.34 \text{ kN}$

Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: User-enteredFactored reaction = $5.21 \times 1.55 = 8.07 \text{ kN}$

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Beam: T8**Span: 6.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.45	0		L	1.35	1.35
U T	Floor	2*.4	0		L	2.40	2.40
R T	Partition	.35*2.5	1		5	1.75	1.75
V T	Partition	.35*1.4	0	.35*2.5	1	0.62	0.06
P T	T12	6.78	4.4			1.81	4.97
						7.93	10.53

Total load: 18.46 kN

Load types: U: UDL R: Part UDL V: Variable load P: PL T: Total (positions in m. from R1)

Maximum B.M. = 15.45 kNm at 3.82 m. from R1

Maximum S.F. = -10.53 kN at R2

Total deflection = $57.0 \times 10^8 / EI$ at 3.10 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 133 x 25 UB S355

$D=203.2$ mm $B=133.2$ mm $t=5.7$ mm $T=7.8$ mm $I_x=2,340$ cm⁴ $r_y=3.10$ cm $Z_x=230$ cm³

Bending

$L_E/r_y = 6.00 \times 100/3.10 = 194$ $D/T = 26.1$

Permissible bending stress, $p_{bc} = 80.9$ N/mm² (Table 3b)

Actual bending stress, $f_{bc} = 15.45 \times 1000/230 = 67.2$ N/mm² OK

Shear

Maximum shear in web, $f_s = 10.53 \times 1000/(5.7 \times 203.2) = 9.1$ N/mm² OK

Beam web

Check unstiffened web capacities with loads of 7.928 kN and 10.53 kN

Bearing: $p_b = 260$ N/mm² (Table 9); $C1 = 39.5$ kN; $C2 = 1.48$ kN/mm

Buckling: $p_c = 181$ N/mm² (Table 17b); $C1 = 105$ kN; $C2 = 1.03$ kN/mm

R1: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 39.5$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 105$ kN

R2: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 39.5$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 105$ kN

Deflection

Total deflection = $57.0 \times 1e8/(205,000 \times 2,340) = 11.9$ mm ($L/505$) OK

Combined bending and shear check (14.c)

Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.690$ at 3.84 m. (≤ 1.25 OK) [14.c]

Bearings**R1: User-entered**

Factored reaction = $7.93 \times 1.55 = 12.29$ kN

Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: User-entered

Factored reaction = $10.53 \times 1.55 = 16.33$ kN

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Beam: T7**Span: 6.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.75	0.75
U T	Floor	2*.4	0		L	2.40	2.40
R T	Partition	1.1*2.5	2.7		6	2.50	6.58
P T	T13	1.64	2.7			0.90	0.74
						<u>6.55</u>	<u>10.47</u>

Total load: 17.01 kN

Load types: U: UDL R: Part UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 14.42 kNm at 3.25 m. from R1

Maximum S.F. = -10.47 kN at R2

Total deflection = $51.9 \times 10^{-8}/EI$ at 3.10 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 102 x 23 UB S355D=203.2 mm B=101.8 mm t=5.4 mm T=9.3 mm $I_x=2,110 \text{ cm}^4$ $r_y=2.36 \text{ cm}$ $Z_x=207 \text{ cm}^3$ **Bending** $L_E/r_y = 6.00 \times 100/2.36 = 254$ D/T = 21.8Permissible bending stress, $p_{bc} = 70.1 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 14.42 \times 1000/207 = 69.6 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 10.47 \times 1000/(5.4 \times 203.2) = 9.5 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 6.548 kN and 10.47 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 41.1 \text{ kN}$; $C2 = 1.40 \text{ kN/mm}$ Buckling: $p_c = 178 \text{ N/mm}^2$ (Table 17b); $C1 = 97.8 \text{ kN}$; $C2 = 0.963 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 41.1 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 97.8 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 41.1 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 97.8 \text{ kN}$ **Deflection**Total deflection = $51.9 \times 10^{-8}/(205,000 \times 2,110) = 12.0 \text{ mm}$ ($L/500$) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.988$ at 3.24 m. (≤ 1.25 OK) [14.c]**Bearings****R1: User-entered**Factored reaction = $6.55 \times 1.55 = 10.15 \text{ kN}$

Web of B2 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: User-enteredFactored reaction = $10.47 \times 1.55 = 16.22 \text{ kN}$

Web of B1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Beam: FJ1**Span: 6.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.30	0.30
R T	Floor	2*.2	0		4.2	1.09	0.59
R T	Floor	2.5*.2	4.2		6	0.14	0.77
P T	Partition	.35*2.5	4.2			0.26	0.61
						1.79	2.27

Total load: 4.06 kN

Load types: U: UDL R: Part UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 3.20 kNm at 3.58 m. from R1

Maximum S.F. = -2.27 kN at R2

Total deflection = $11.9 \times 10^8 / EI$ at 3.10 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 75 x 220 C24 6.9 kg/m approx $z = 605.0 \text{ cm}^3$ $I = 6,655 \text{ cm}^4$ Timber grade: C24 Load sharing system: $K_8 = 1.1$ [§2.10.11] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 10,800 \text{ N/mm}^2$ (E_{mean})**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 3.20 \times 1000 / 605 = 5.29 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 2.27 \times 1000 \times 3 / (2 \times 75 \times 220) = 0.21 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $11.91 \times 10^8 / (10,800 \times 6,655) = 16.57 \text{ mm}$ Mid-span shear deflection = $1.2 \times 3.12 \times 10^6 / ((E/16) \times 75 \times 220) = 0.34 \text{ mm}$ Total deflection = $16.57 + 0.34 = 16.91 \text{ mm}$ (0.0028 L) $\leq 0.003L$ OK

Beam: FJ2**Span: 6.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.30	0.30
U T	Floor	2*.2	0		L	1.20	1.20
						1.50	1.50

Total load: 3.00 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 2.25 kNm at 3.00 m. from R1

Maximum S.F. = 1.50 kN at 0.00 m. from R1

Total deflection = $8.44 \times 10^8 / EI$ at 3.00 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 50 x 220 C24 4.6 kg/m approx $z = 403.3 \text{ cm}^3$ $I = 4,437 \text{ cm}^4$ Timber grade: C24 Load sharing system: $K_8 = 1.1$ [§2.10.11] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 10,800 \text{ N/mm}^2$ (E_{mean})**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 2.25 \times 1000/403 = 5.58 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 1.50 \times 1000 \times 3/(2 \times 50 \times 220) = 0.20 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $8.44 \times 10^8 / (10,800 \times 4,437) = 17.61 \text{ mm}$ Mid-span shear deflection = $1.2 \times 2.25 \times 10^6 / ((E/16) \times 50 \times 220) = 0.36 \text{ mm}$ Total deflection = $17.61 + 0.36 = 17.97 \text{ mm}$ (0.0030 L) $\leq 0.003L$ OK

Beam: T9**Span: 1.8 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.09	0.09
U T	Stairs	1.5	0		L	1.35	1.35
U T	Floor	2*1.1/2	0		L	0.99	0.99
						2.43	2.43

Total load: 4.86 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 1.09 kNm at 0.90 m. from R1

Maximum S.F. = 2.43 kN at 0.00 m. from R1

Total deflection = $0.369 \times 10^8 / EI$ at 0.90 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 2no 47 x 220 C24 8.7 kg/m approx $z = 758.3 \text{ cm}^3$ $I = 8,341 \text{ cm}^4$ Timber grade: C24 2 members acting together: $K_8 = 1.1$ [§2.9] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 7,200 \times 1.14 = 8,208 \text{ N/mm}^2$ ($E_{min} \cdot K_9$)**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 1.09 \times 1000 / 758 = 1.44 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,II} = t_{g,II} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 2.43 \times 1000 \times 3 / (2 \times 94 \times 220) = 0.18 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.369 \times 10^8 / (8,208 \times 8,341) = 0.54 \text{ mm}$ Mid-span shear deflection = $1.2 \times 1.09 \times 10^6 / ((E/16) \times 94 \times 220) = 0.12 \text{ mm}$ Total deflection = $0.54 + 0.12 = 0.66 \text{ mm}$ ($0.0004 L$) $\leq 0.003L$ OK

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Beam: T10**Span: 0.9 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.04	0.04
U T	Stairs	1.5	0		L	0.67	0.67
U T	Floor	2*1.7/2	0		L	0.76	0.76
						1.48	1.48

Total load: 2.97 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 0.33 kNm at 0.45 m. from R1

Maximum S.F. = 1.48 kN at 0.00 m. from R1

Total deflection = $0.028 \times 10^8 / EI$ at 0.45 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 2no 47 x 220 C24 8.7 kg/m approx $z = 758.3 \text{ cm}^3$ $I = 8,341 \text{ cm}^4$ Timber grade: C24 2 members acting together: $K_8 = 1.1$ [§2.9] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 7,200 \times 1.14 = 8,208 \text{ N/mm}^2$ ($E_{\min} \cdot K_9$)**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 0.334 \times 1000/758 = 0.44 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 1.48 \times 1000 \times 3/(2 \times 94 \times 220) = 0.11 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.028 \times 10^8 / (8,208 \times 8,341) = 0.04 \text{ mm}$ Mid-span shear deflection = $1.2 \times 0.334 \times 10^6 / ((E/16) \times 94 \times 220) = 0.04 \text{ mm}$ Total deflection = $0.04 + 0.04 = 0.08 \text{ mm}$ ($0.0001 L$) $\leq 0.003L$ OK**Bearings****R1: User-entered**Stress on underside of timber at bearing = $1.48 \times 1000 / (94 \times 100) = 0.158 \text{ N/mm}^2$ Permissible stress = $s_{c,g,i} \cdot K_3 = 2.40 \times 1.00 = 2.40 \text{ N/mm}^2$ OKFactored reaction = $1.48 \times 1.55 = 2.30 \text{ kN}$

Hanger to T5

R2: User-enteredStress on underside of timber at bearing = $1.48 \times 1000 / (94 \times 100) = 0.158 \text{ N/mm}^2$ Permissible stress = $s_{c,g,i} \cdot K_3 = 2.40 \times 1.00 = 2.40 \text{ N/mm}^2$ OKFactored reaction = $1.48 \times 1.55 = 2.30 \text{ kN}$

Post to floor

Beam: T11**Span: 2.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.10	0.10
U T	Floor	2*1.2/2	0		L	1.20	1.20
R T	Baluster	.35*1.1	.9		2	0.12	0.31
						1.42	1.61

Total load: 3.02 kN

Load types: U: UDL R: Part UDL T: Total (positions in m. from R1)

Maximum B.M. = 0.77 kNm at 1.05 m. from R1

Maximum S.F. = -1.61 kN at R2

Total deflection = $0.317 \times 10^8 / EI$ at 1.02 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 2no 47 x 220 C24 8.7 kg/m approx $z = 758.3 \text{ cm}^3$ $I = 8,341 \text{ cm}^4$ Timber grade: C24 2 members acting together: $K_8 = 1.1$ [§2.9] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 7,200 \times 1.14 = 8,208 \text{ N/mm}^2$ ($E_{\min} \cdot K_9$)**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 0.766 \times 1000/758 = 1.01 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,||} = t_{g,||} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 1.61 \times 1000 \times 3/(2 \times 94 \times 220) = 0.12 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.317 \times 10^8 / (8,208 \times 8,341) = 0.46 \text{ mm}$ Mid-span shear deflection = $1.2 \times 0.765 \times 10^6 / ((E/16) \times 94 \times 220) = 0.09 \text{ mm}$ Total deflection = $0.46 + 0.09 = 0.55 \text{ mm}$ ($0.0003 L$) $\leq 0.003L$ OK**Bearings** (bearing plate sizing to BS5950-1:2000)

Factor reactions by 1.55 (user selected value)

R1: User-enteredStress on underside of timber at bearing = $1.42 \times 1000 / (94 \times 100) = 0.151 \text{ N/mm}^2$ Permissible stress = $s_{c,g,i} \cdot K_3 = 2.40 \times 1.00 = 2.40 \text{ N/mm}^2$ OKFactored reaction = $1.42 \times 1.55 = 2.20 \text{ kN}$

hanger to T11

R2: 94 x 100 mm bearing plateStress on underside of timber at bearing = $1.61 \times 1000 / (94 \times 100) = 0.171 \text{ N/mm}^2$ Permissible stress = $s_{c,g,i} \cdot K_3 = 2.40 \times 1.00 = 2.40 \text{ N/mm}^2$ OKFactored reaction = $1.61 \times 1.55 = 2.49 \text{ kN}$

10 mm m.s. bearing plate, size 94 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)Bearing plate projection = $94/2 = 47 \text{ mm}$

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Factored stress under plate = $1.55 \times 1.61 \times 1000/94 \times 100 = 0.26 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.26 \times 47.0 \times 47.0/275)} = 2.53 \text{ mm}$: use 10mm

Factored bending stress in plate = $0.26 \times 47.0 \times (47.0/2)/(10 \times 10/6) = 17.6 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: T12**Span: 2.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.10	0.10
U T	Partition	.35*2.5	0		L	0.88	0.88
U T	Floor	2*1.3/2	0		L	1.30	1.30
U T	Floor	2*4.5/2	0		L	4.50	4.50
						<u>6.78</u>	<u>6.78</u>

Total load: 13.55 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 3.39 kNm at 1.00 m. from R1

Maximum S.F. = 6.77 kN at 0.00 m. from R1

Total deflection = $1.41 \times 10^8 / EI$ at 1.00 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 2no 47 x 220 C24 8.7 kg/m approx $z = 758.3 \text{ cm}^3$ $I = 8,341 \text{ cm}^4$ Timber grade: C24 2 members acting together: $K_8 = 1.1$ [§2.9] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 7,200 \times 1.14 = 8,208 \text{ N/mm}^2$ ($E_{\min} \cdot K_9$)**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 3.39 \times 1000/758 = 4.47 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,II} = t_{g,II} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 6.77 \times 1000 \times 3/(2 \times 94 \times 220) = 0.49 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $1.41 \times 10^8 / (8,208 \times 8,341) = 2.06 \text{ mm}$ Mid-span shear deflection = $1.2 \times 3.39 \times 10^6 / ((E/16) \times 94 \times 220) = 0.38 \text{ mm}$ Total deflection = $2.06 + 0.38 = 2.44 \text{ mm}$ ($0.0012 L$) $\leq 0.003L$ OK**Bearings** (bearing plate sizing to BS5950-1:2000)

Factor reactions by 1.55 (user selected value)

R1: User-enteredStress on underside of timber at bearing = $6.78 \times 1000 / (94 \times 100) = 0.721 \text{ N/mm}^2$ Permissible stress = $s_{c,g,i} \cdot K_3 = 2.40 \times 1.00 = 2.40 \text{ N/mm}^2$ OKFactored reaction = $6.78 \times 1.55 = 10.50 \text{ kN}$

Hangere to T8

R2: 175 x 100 mm bearing plateStress on underside of timber at bearing = $6.78 \times 1000 / (94 \times 100) = 0.721 \text{ N/mm}^2$ Permissible stress = $s_{c,g,i} \cdot K_3 = 2.40 \times 1.00 = 2.40 \text{ N/mm}^2$ OKFactored reaction = $6.78 \times 1.55 = 10.50 \text{ kN}$

10 mm m.s. bearing plate, size 175 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)Bearing plate projection = $175/2 = 87.5 \text{ mm}$

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Factored stress under plate = $1.55 \times 6.78 \times 1000/175 \times 100 = 0.60 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.60 \times 87.5 \times 87.5/275)} = 7.08 \text{ mm}$: use 10mm

Factored bending stress in plate = $0.60 \times 87.5 \times (87.5/2)/(10 \times 10/6) = 137.8 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: T13**Span: 1.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.1	0		L	0.05	0.05
U T	Partition	.35*2.5	0		L	0.44	0.44
U T	Floor	2*2.3/2	0		L	1.15	1.15
						<u>1.64</u>	<u>1.64</u>

Total load: 3.28 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 0.41 kNm at 0.50 m. from R1

Maximum S.F. = 1.64 kN at 0.00 m. from R1

Total deflection = $0.043 \times 10^8/EI$ at 0.50 m. from R1 (E in N/mm^2 , I in cm^4)

Timber beam calculation to BS5268 Part 2: 2002 using C24 timber

Use 2no 50 x 220 C24 9.2 kg/m approx $z = 806.7 \text{ cm}^3$ $I = 8,873 \text{ cm}^4$ Timber grade: C24 2 members acting together: $K_8 = 1.1$ [§2.9] K_3 (loading duration factor) = 1.00 (long term) K_7 (depth factor) = $(300/220)^{0.11} = 1.03$ [§2.10.6] K_8 (load sharing factor) = 1.1 [§2.9,2.10] $E = 7,200 \times 1.14 = 8,208 \text{ N/mm}^2$ ($E_{\min} \cdot K_9$)**Bending**Permissible bending stress, $s_{m,adm} = s_{m,g} \cdot K_3 \cdot K_7 \cdot K_8 = 7.5 \times 1.00 \times 1.03 \times 1.1 = 8.54 \text{ N/mm}^2$ Applied bending stress, $s_{m,a} = 0.409 \times 1000/807 = 0.51 \text{ N/mm}^2$ OK**Shear**Permissible shear stress, $t_{adm,II} = t_{g,II} \cdot K_3 \cdot K_8 = 0.71 \times 1.00 \times 1.1 = 0.78 \text{ N/mm}^2$ Applied shear stress, $t_a = 1.64 \times 1000 \times 3/(2 \times 100 \times 220) = 0.11 \text{ N/mm}^2$ OK**Deflection**Bending deflection = $0.043 \times 10^8/(8,208 \times 8,873) = 0.06 \text{ mm}$ Mid-span shear deflection = $1.2 \times 0.409 \times 10^6/((E/16) \times 100 \times 220) = 0.04 \text{ mm}$ Total deflection = $0.06 + 0.04 = 0.10 \text{ mm}$ ($0.0001 L$) $\leq 0.003L$ OK

Beam: B2**Span: 6.4 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.7	0		L	2.24	2.24
R T	Floor	2*2.7	.4		1.7	5.87	1.15
R T	Floor	1.79/0.4	1.7		3.8	5.36	4.04
R T	Floor	2*6/2	3.8		4.8	1.97	4.03
R T	Floor	2*4.8/2	4.8		6.4	0.96	6.72
R T	Partition	.35*1.4	.4		6.1	1.37	1.42
R T	Eaves	1.7*1/2	.4		6.1	2.38	2.46
R T	Pitched roof	1.7*1.4/2	.4		6.1	3.34	3.44
P T	Beam: T6 : R1	8.61 [B/F]	.4			8.07	0.54
P T	Beam: T7 : R1	6.55 [B/F]	1.6			4.91	1.64
P T	Beam: T8 : R1	7.93 [B/F]	4.8			1.98	5.95
						38.46	33.63

Total load: 72.08 kN

Load types: U: UDL R: Part UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 54.86 kNm at 3.23 m. from R1

Maximum S.F. = 38.46 kN at 0.00 m. from R1

Total deflection = $241 \times 10^8 / EI$ at 3.20 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 203 x 71 UC S355

$D=215.8$ mm $B=206.4$ mm $t=10.0$ mm $T=17.3$ mm $I_x=7,620$ cm⁴ $r_y=5.30$ cm $Z_x=706$ cm³

Bending

$$L_E/r_y = 6.40 \times 100/5.30 = 121 \quad D/T = 12.5$$

Permissible bending stress, $p_{bc} = 174.1$ N/mm² (Table 3b)

Actual bending stress, $f_{bc} = 54.9 \times 1000/706 = 77.7$ N/mm² OK

Shear

Maximum shear in web, $f_s = 38.46 \times 1000/(10.0 \times 215.8) = 17.8$ N/mm² OK

Beam web

Check unstiffened web capacities with loads of 38.46 kN and 33.63 kN

Bearing: $p_b = 260$ N/mm² (Table 9); $C1 = 124$ kN; $C2 = 2.60$ kN/mm

Buckling: $p_c = 201$ N/mm² (Table 17b); $C1 = 217$ kN; $C2 = 2.01$ kN/mm

R1: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 124$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 217$ kN

R2: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 124$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 217$ kN

Deflection

Total deflection = $241 \times 1e8/(205,000 \times 7,620) = 15.4$ mm ($L/415$) OK

Combined bending and shear check (14.c)

Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.199$ at 3.20 m. (≤ 1.25 OK) [14.c]

Bearings (bearing plate sizing to BS5950-1:2000)

203 x 203 x 71 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 60.9 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-entered

Factored reaction = $38.46 \times 1.55 = 59.6 \text{ kN}$

Web of T1 via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: 650 x 120 mm bearing plate

Factored reaction = $33.63 \times 1.55 = 52.1 \text{ kN}$

30 mm m.s. bearing plate, size 650 x 120 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(650 - 60.9)/2 = 294.5 \text{ mm}$

Factored stress under plate = $1.55 \times 33.63 \times 1000 / (650 \times 120) = 0.67 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.67 \times 295 \times 295 / 265)} = 25.6 \text{ mm}$: use 30mm

Factored bending stress in plate = $0.67 \times 295 \times (295/2) / (30 \times 30/6) = 193.2 \text{ N/mm}^2$ ($p_y = 265 \text{ N/mm}^2$)

Beam: B1**Span: 6.8 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.75	0		L	2.55	2.55
R T	Floor	2.27/0.4	2		3.8	5.86	4.36
R T	Floor	2*6/2	3.8		4.8	2.21	3.79
R T	Floor	2*1.4/2	4.8		6.8	0.41	2.39
R T	Floor	2*2.9/2	0		1.9	4.74	0.77
R T	Floor	2*4/2	1.9		4.7	5.76	5.44
R T	Floor	2*.8/2	4.8		6.8	0.24	1.36
P T	Beam: T2 : R1	20.31 [B/F]	.4			19.11	1.19
P T	Beam: T1 : R2	11.09 [B/F]	.4			10.44	0.65
P T	Beam: T6 : R2	5.21 [B/F]	.9			4.52	0.69
P T	Beam: T7 : R2	10.47 [B/F]	1.9			7.54	2.92
P T	Beam: T4 : R1	4.53 [B/F]	1.9			3.27	1.27
P T	Beam: T5 : R1	4.11 [B/F]	4.8			1.21	2.90
P T	Beam: T8 : R2	10.53 [B/F]	4.8			3.10	7.44
						70.95	37.72

Total load: 108.67 kN

Load types: U: UDL R: Part UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 83.96 kNm at 3.14 m. from R1

Maximum S.F. = 70.95 kN at 0.00 m. from R1

Total deflection = $411 \times 10^8 / EI$ at 3.34 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 254 x 254 x 73 UC S355

$D=254.1$ mm $B=254.6$ mm $t=8.6$ mm $T=14.2$ mm $I_x=11,400$ cm⁴ $r_y=6.48$ cm $Z_x=898$ cm³

Bending

$L_E/r_y = 6.80 \times 100/6.48 = 105$ $D/T = 17.9$

Permissible bending stress, $p_{bc} = 167.0$ N/mm² (Table 3b)

Actual bending stress, $f_{bc} = 84.0 \times 1000/898 = 93.5$ N/mm² OK

Shear

Maximum shear in web, $f_s = 71.0 \times 1000/(8.6 \times 254.1) = 32.5$ N/mm² OK

Beam web

Check unstiffened web capacities with loads of 70.95 kN and 37.72 kN

Bearing: $p_b = 260$ N/mm² (Table 9); $C1 = 104$ kN; $C2 = 2.24$ kN/mm

Buckling: $p_c = 193$ N/mm² (Table 17b); $C1 = 211$ kN; $C2 = 1.66$ kN/mm

R1: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 104$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 211$ kN

R2: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 104$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 211$ kN

Deflection

Total deflection = $411 \times 1e8/(205,000 \times 11,400) = 17.6$ mm ($L/387$) OK

Combined bending and shear check (14.c)

Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.314$ at 3.13 m. (≤ 1.25 OK) [14.c]

Bearings (bearing plate sizing to BS5950-1:2000)

254 x 254 x 73 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 57.3$ mm

Factor reactions by 1.55 (user selected value)

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

R1: 1600 x 100 mm padstone

Factored reaction = $70.95 \times 1.55 = 110.0 \text{ kN}$

Factored stress under padstone = $1.55 \times 70.95 \times 1000/1600 \times 100 = 0.69 \text{ N/mm}^2$

Or shs

R2: 850 x 100 mm bearing plate

Factored reaction = $37.72 \times 1.55 = 58.5 \text{ kN}$

40 mm m.s. bearing plate, size 850 x 100 mm

Bearing plate projection beyond stiff bearing length = $(850 - 57.3)/2 = 396.3 \text{ mm}$

Factored stress under plate = $1.55 \times 37.72 \times 1000/850 \times 100 = 0.69 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.69 \times 396 \times 396/265)} = 35.0 \text{ mm}$: use 40mm

Factored bending stress in plate = $0.69 \times 396 \times (396/2)/(40 \times 40/6) = 202.6 \text{ N/mm}^2$ ($p_y = 265 \text{ N/mm}^2$)

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Beam: l1**Span: 1.1 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.5	0		L	0.28	0.28
U T	Wall	5*1	0		L	2.75	2.75
U T	Eaves	1.7*.3/2	0		L	0.14	0.14
U T	Dormer face	1.1*1.7	0		L	1.03	1.03
U T	Flat roof	1.5*2/2	0		L	0.83	0.83
						5.02	5.02

Total load: 10.04 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 1.38 kNm at 0.55 m. from R1

Maximum S.F. = 5.02 kN at 0.00 m. from R1

Total deflection = $0.174 \times 10^8 / EI$ at 0.55 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 203 x 46 UC S355D=203.2 mm B=203.6 mm t=7.2 mm T=11.0 mm $I_x=4,570 \text{ cm}^4$ $r_y=5.13 \text{ cm}$ $Z_x=450 \text{ cm}^3$ **Bending** $L_E/r_y = 1.10 \times 100/5.13 = 21$ D/T = 18.5Permissible bending stress, $p_{bc} = 230.0 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 1.38 \times 1000/450 = 3.07 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 5.02 \times 1000/(7.2 \times 203.2) = 3.4 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacity with load of 5.019 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 68.7 \text{ kN}$; $C2 = 1.87 \text{ kN/mm}$ Buckling: $p_c = 194 \text{ N/mm}^2$ (Table 17b); $C1 = 142 \text{ kN}$; $C2 = 1.40 \text{ kN/mm}$ Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 68.7 \text{ kN} < < <$ Buckling capacity, $P_x = C1 + L_b.C2 = 142 \text{ kN}$ **Deflection**Total deflection = $0.174 \times 1e8/(205,000 \times 4,570) = 0.0 \text{ mm}$ OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.001$ at 0.00 m. (≤ 1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)203 x 203 x 46 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 45.5 \text{ mm}$

Factor reactions by 1.55 (user selected value)

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

R1: 204 x 205 mm bearing plate

Factored reaction = $5.02 \times 1.55 = 7.78 \text{ kN}$

10 mm m.s. bearing plate, size 204 x 205 mm

Bearing plate projection beyond stiff bearing length = $(204 - 45.5)/2 = 79.2 \text{ mm}$

Factored stress under plate = $1.55 \times 5.02 \times 1000 / (204 \times 205) = 0.19 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.19 \times 79.2 \times 79.2 / 275)} = 3.57 \text{ mm}$: use 10 mm

Factored bending stress in plate = $0.19 \times 79.2 \times (79.2/2) / (10 \times 10^6) = 35.0 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

R2 as R1

Beam: BA**Span: 5.2 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.65	0.65
U T	Floor	2*.4	0		L	2.08	2.08
U T	Partition	.45*3.2	0		L	3.74	3.74
						<u>6.47</u>	<u>6.47</u>

Total load: 12.95 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 8.42 kNm at 2.60 m. from R1

Maximum S.F. = 6.47 kN at 0.00 m. from R1

Total deflection = $23.7 \times 10^8 / EI$ at 2.60 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 178 x 102 x 19 UB S355D=177.8 mm B=101.2 mm t=4.8 mm T=7.9 mm $I_x=1,360 \text{ cm}^4$ $r_y=2.37 \text{ cm}$ $Z_x=153 \text{ cm}^3$ **Bending** $L_E/r_y = 5.20 \times 100/2.37 = 219$ D/T = 22.5Permissible bending stress, $p_{bc} = 79.2 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 8.42 \times 1000/153 = 55.0 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 6.47 \times 1000/(4.8 \times 177.8) = 7.6 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacity with load of 6.474 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 33.5 \text{ kN}$; $C2 = 1.25 \text{ kN/mm}$ Buckling: $p_c = 180 \text{ N/mm}^2$ (Table 17b); $C1 = 76.8 \text{ kN}$; $C2 = 0.864 \text{ kN/mm}$ Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 33.5 \text{ kN} < < <$ Buckling capacity, $P_x = C1 + L_b.C2 = 76.8 \text{ kN}$ **Deflection**Total deflection = $23.7 \times 10^8 / (205,000 \times 1,360) = 8.5 \text{ mm}$ (L/612) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.483$ at 2.60 m. (≤ 1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)178 x 102 x 19 UB stiff bearing length, $b_1 = t + 1.6r + 2T = 32.8 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-enteredFactored reaction = $6.47 \times 1.55 = 10.03 \text{ kN}$

Web of Lower spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: 150 x 100 mm bearing plate

Factored reaction = $6.47 \times 1.55 = 10.03 \text{ kN}$

10 mm m.s. bearing plate, size 150 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(150 - 32.8)/2 = 58.6 \text{ mm}$

Factored stress under plate = $1.55 \times 6.47 \times 1000 / 150 \times 100 = 0.67 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.67 \times 58.6 \times 58.6 / 275)} = 5.01 \text{ mm}$: use 10mm

Factored bending stress in plate = $0.67 \times 58.6 \times (58.6/2) / (10 \times 10/6) = 69.0 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: BB**Span: 4.1 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.51	0.51
U T	Floor	2*.4	0		L	1.64	1.64
U T	Partition	.45*3.2	0		L	2.95	2.95
						5.10	5.10

Total load: 10.21 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 5.23 kNm at 2.05 m. from R1

Maximum S.F. = 5.10 kN at 0.00 m. from R1

Total deflection = $9.16 \times 10^{-8}/EI$ at 2.05 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 178 x 102 x 19 UB S355D=177.8 mm B=101.2 mm t=4.8 mm T=7.9 mm $I_x=1,360 \text{ cm}^4$ $r_y=2.37 \text{ cm}$ $Z_x=153 \text{ cm}^3$ **Bending** $L_E/r_y = 4.10 \times 100/2.37 = 173$ D/T = 22.5Permissible bending stress, $p_{bc} = 98.0 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 5.23 \times 1000/153 = 34.2 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 5.10 \times 1000/(4.8 \times 177.8) = 6.0 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacity with load of 5.104 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 33.5 \text{ kN}$; $C2 = 1.25 \text{ kN/mm}$ Buckling: $p_c = 180 \text{ N/mm}^2$ (Table 17b); $C1 = 76.8 \text{ kN}$; $C2 = 0.864 \text{ kN/mm}$ Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 33.5 \text{ kN} < < <$ Buckling capacity, $P_x = C1 + L_b.C2 = 76.8 \text{ kN}$ **Deflection**Total deflection = $9.16 \times 1e8/(205,000 \times 1,360) = 3.3 \text{ mm}$ (L/1248) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.122$ at 2.05 m. (≤ 1.25 OK) [14.c]**Bearings (bearing plate sizing to BS5950-1:2000)**178 x 102 x 19 UB stiff bearing length, $b_1 = t + 1.6r + 2T = 32.8 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: 125 x 100 mm bearing plateFactored reaction = $5.10 \times 1.55 = 7.91 \text{ kN}$

10 mm m.s. bearing plate, size 125 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)Bearing plate projection beyond stiff bearing length = $(125-32.8)/2 = 46.1 \text{ mm}$ Factored stress under plate = $1.55 \times 5.10 \times 1000/125 \times 100 = 0.63 \text{ N/mm}^2$ Required plate thickness = $\sqrt{(3 \times 0.63 \times 46.1 \times 46.1/275)} = 3.83 \text{ mm}$: use 10mmFactored bending stress in plate = $0.63 \times 46.1 \times (46.1/2)/(10 \times 10/6) = 40.4 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)**R2: User-entered**

MS

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Factored reaction = $5.10 \times 1.55 = 7.91 \text{ kN}$

Web of Lower spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

Beam: Lower Ground Spine beam**Span: 6.6 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.9	0		L	2.97	2.97
U T	Floor	2.79/0.4	0		L	23.02	23.02
U T	Floor	2.63/0.4	0		L	21.70	21.70
P T	Partition	4.53	1.8			3.29	1.24
P T	Partition	4.53	3.5			2.13	2.40
P T	Beam: BA : R1	6.47 [B/F]	4.7			1.86	4.61
P T	Beam: BB : R2	5.10 [B/F]	5			1.24	3.87
						56.21	59.80

Total load: 116.01 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 100.52 kNm at 3.50 m. from R1

Maximum S.F. = -59.80 kN at R2

Total deflection = $454 \times 10^8 / EI$ at 3.36 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 254 x 254 x 89 UC S355D=260.3 mm B=256.3 mm t=10.3 mm T=17.3 mm $I_x=14,300 \text{ cm}^4$ $r_y=6.55 \text{ cm}$ $Z_x=1,100 \text{ cm}^3$ **Bending** $L_E/r_y = 6.60 \times 100/6.55 = 101$ D/T = 15.0Permissible bending stress, $p_{bc} = 179.1 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 100.5 \times 1000/1,100 = 91.4 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 59.8 \times 1000/(10.3 \times 260.3) = 22.3 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacities with loads of 56.21 kN and 59.80 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 139 \text{ kN}$; $C2 = 2.68 \text{ kN/mm}$ Buckling: $p_c = 197 \text{ N/mm}^2$ (Table 17b); $C1 = 265 \text{ kN}$; $C2 = 2.03 \text{ kN/mm}$ R1: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 139 \text{ kN} \lll$ Buckling capacity, $P_x = C1 + L_b.C2 = 265 \text{ kN}$ R2: Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 139 \text{ kN} \lll$ Buckling capacity, $P_x = C1 + L_b.C2 = 265 \text{ kN}$ **Deflection**Total deflection = $454 \times 1e8/(205,000 \times 14,300) = 15.5 \text{ mm}$ ($L/426$) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.260$ at 3.43 m. (≤ 1.25 OK) [14.c]**Bearings**254 x 254 x 89 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 65.2 \text{ mm}$

Factor reactions by 1.55 (user selected value)

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

R1: 1050 x 120 mm padstone

Factored reaction = $56.21 \times 1.55 = 87.1 \text{ kN}$

Factored stress under padstone = $1.55 \times 56.21 \times 1000 / (1050 \times 120) = 0.69 \text{ N/mm}^2$

Or SHS

R2: 1150 x 120 mm padstone

Factored reaction = $59.80 \times 1.55 = 92.7 \text{ kN}$

Factored stress under padstone = $1.55 \times 59.80 \times 1000 / (1150 \times 120) = 0.67 \text{ N/mm}^2$

Or SHS

Beam: SHS1(R1 of lower ground floor beam)**Span: 1.0 m.**

Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T o.w.	0.45	0		L	0.22	0.22
P T Beam: Lower Ground : R1	56.21 [B/F]	.5			28.10	28.10
					28.33	28.33

Total load: 56.66 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 14.11 kNm at 0.50 m. from R1

Maximum S.F. = 28.33 kN at 0.00 m. from R1

Total deflection = $1.18 \times 10^8 / EI$ at 0.50 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 120 x 120 x 12.5 HF SHS S355 (40.9kg/m) $I_x = 982 \text{ cm}^4$ $r_y = 4.34 \text{ cm}$ $Z_x = 164 \text{ cm}^3$ $A = 52.1 \text{ cm}^2$ **Bending** $L_E/r_y = 1.00 \times 100/4.34 = 23$ $D/T = 9.6$ Permissible bending stress, $p_{bc} = 230.0 \text{ N/mm}^2$ (Table 2)Actual bending stress, $f_{bc} = 14.11 \times 1000/164 = 86.0 \text{ N/mm}^2$ OK**Shear**Section shear area = $0.5 \times CSA = 52.1 \times 100 \times 0.5 = 2,605 \text{ mm}^2$ (clause 23)Maximum shear stress, $f_s = 28.33 \times 1000/2,605 = 10.9 \text{ N/mm}^2$ OK

Web buckling and crushing have not been checked

DeflectionTotal deflection = $1.18 \times 10^8 / (205,000 \times 982) = 0.6 \text{ mm}$ ($L/1711$) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.146$ at 0.50 m. (≤ 1.25 OK) [14.c]

Beam: SHS2(R2 of Lower ground floor beam)**Span: 1.0 m.**

Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T o.w.	0.45	0		L	0.22	0.22
P T Beam: Lower Ground : R2	59.80 [B/F]	.5			29.90	29.90
					30.13	30.13

Total load: 60.25 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 15.01 kNm at 0.50 m. from R1

Maximum S.F. = 30.13 kN at 0.00 m. from R1

Total deflection = $1.25 \times 10^8 / EI$ at 0.50 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 120 x 120 x 12.5 HF SHS S355 (40.9kg/m) $I_x = 982 \text{ cm}^4$ $r_y = 4.34 \text{ cm}$ $Z_x = 164 \text{ cm}^3$ $A = 52.1 \text{ cm}^2$ **Bending** $L_E / r_y = 1.00 \times 100 / 4.34 = 23$ $D/T = 9.6$ Permissible bending stress, $p_{bc} = 230.0 \text{ N/mm}^2$ (Table 2)Actual bending stress, $f_{bc} = 15.01 \times 1000 / 164 = 91.5 \text{ N/mm}^2$ OK**Shear**Section shear area = $0.5 \times CSA = 52.1 \times 100 \times 0.5 = 2,605 \text{ mm}^2$ (clause 23)Maximum shear stress, $f_s = 30.13 \times 1000 / 2,605 = 11.6 \text{ N/mm}^2$ OK

Web buckling and crushing have not been checked

DeflectionTotal deflection = $1.25 \times 1e8 / (205,000 \times 982) = 0.6 \text{ mm}$ ($L/1608$) OK**Combined bending and shear check (14.c)**Check $(f_{bc} / p_{bc})^2 + (f_s / p_s)^2 = 0.165$ at 0.50 m. (≤ 1.25 OK) [14.c]

Beam: Upper Ground floor spine beam**Span: 6.6 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	1.0	0		L	3.30	3.30
R T	Floor	4.45/0.4	0		4.7	33.67	18.62
R T	Floor	3.99/0.4	4.7		6.6	2.73	16.22
R T	Floor	3.6/0.4	0		4.7	27.24	15.06
R T	Floor	4.8/0.4	4.7		6.6	3.28	19.52
P T	Partition	5.74	4.7			1.65	4.09
P T	Partition	5.5	4.7			1.58	3.92
P T	Partition	2.8	3.6			1.27	1.53
P T	BC	5.09	4.7			1.47	3.62
P T	BD	5.7	4.7			1.64	4.06
						77.83	89.94

Total load: 167.77 kN

Load types: U: UDL R: Part UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 143.31 kNm at 3.60 m. from R1

Maximum S.F. = -89.94 kN at R2

Total deflection = $649 \times 10^8 / EI$ at 3.36 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 305 x 305 x 97 UC S355

$D=307.9$ mm $B=305.3$ mm $t=9.9$ mm $T=15.4$ mm $I_x=22,300$ cm⁴ $r_y=7.69$ cm $Z_x=1,450$ cm³

Bending

$$L_E/r_y = 6.60 \times 100/7.69 = 86 \quad D/T = 20.0$$

Permissible bending stress, $p_{bc} = 183.0$ N/mm² (Table 3b)

Actual bending stress, $f_{bc} = 143.3 \times 1000/1,450 = 98.8$ N/mm² OK

Shear

Maximum shear in web, $f_s = 89.9 \times 1000/(9.9 \times 307.9) = 29.5$ N/mm² OK

Beam web

Check unstiffened web capacities with loads of 77.83 kN and 89.94 kN

Bearing: $p_b = 260$ N/mm² (Table 9); $C1 = 136$ kN; $C2 = 2.57$ kN/mm

Buckling: $p_c = 191$ N/mm² (Table 17b); $C1 = 291$ kN; $C2 = 1.89$ kN/mm

R1: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 136$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 291$ kN

R2: Minimum required stiff bearing length, $L_b = 0$ mm

Bearing capacity, $P_w = C1 + L_b.C2 = 136$ kN <<<

Buckling capacity, $P_x = C1 + L_b.C2 = 291$ kN

Deflection

Total deflection = $649 \times 1e8/(205,000 \times 22,300) = 14.2$ mm ($L/465$) OK

Combined bending and shear check (14.c)

Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.291$ at 3.56 m. (≤ 1.25 OK) [14.c]

Bearings

305 x 305 x 97 UC stiff bearing length, $b_1 = t + 1.6r + 2T = 65 \text{ mm}$

Factor reactions by 1.55 (user selected value)

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

R1: 1750 x 100 x 75h mm padstone

Factored reaction = $77.83 \times 1.55 = 120.6 \text{ kN}$

Factored stress under padstone = $1.55 \times 77.83 \times 1000 / 1750 \times 100 = 0.69 \text{ N/mm}^2$

Or SHS

R2: 2000 x 100 mm padstone

Factored reaction = $89.94 \times 1.55 = 139.4 \text{ kN}$

Factored stress under padstone = $1.55 \times 89.94 \times 1000 / 2000 \times 100 = 0.70 \text{ N/mm}^2$

Or SHS

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Beam: SHS3 (R1 of B1)**Span: 1.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U	T o.w.	0.45	0		L	0.22	0.22
P	T Beam: B1 : R1	70.95 [B/F]	.5			35.48	35.48
						35.70	35.70

Total load: 71.40 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 17.79 kNm at 0.50 m. from R1

Maximum S.F. = 35.70 kN at 0.00 m. from R1

Total deflection = $1.48 \times 10^8 / EI$ at 0.50 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 120 x 120 x 12.5 HF SHS S355 (40.9kg/m) $I_x = 982 \text{ cm}^4$ $r_y = 4.34 \text{ cm}$ $Z_x = 164 \text{ cm}^3$ $A = 52.1 \text{ cm}^2$ **Bending** $L_E / r_y = 1.00 \times 100 / 4.34 = 23$ $D/T = 9.6$ Permissible bending stress, $p_{bc} = 230.0 \text{ N/mm}^2$ (Table 2)Actual bending stress, $f_{bc} = 17.79 \times 1000 / 164 = 108 \text{ N/mm}^2$ OK**Shear**Section shear area = $0.5 \times CSA = 52.1 \times 100 \times 0.5 = 2,605 \text{ mm}^2$ (clause 23)Maximum shear stress, $f_s = 35.70 \times 1000 / 2,605 = 13.7 \text{ N/mm}^2$ OK

Web buckling and crushing have not been checked

DeflectionTotal deflection = $1.48 \times 1e8 / (205,000 \times 982) = 0.7 \text{ mm}$ ($L/1357$) OK**Combined bending and shear check (14.c)**Check $(f_{bc} / p_{bc})^2 + (f_s / p_s)^2 = 0.232$ at 0.50 m. (≤ 1.25 OK) [14.c]

Beam: SHS4 (R2 of Upper ground floor beam)**Span: 1.0 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U	T o.w.	0.45	0		L	0.22	0.22
P	T Beam: : R2	89.94 [B/F]	.5			44.97	44.97
						45.19	45.19

Total load: 90.39 kN

Load types: U: UDL P: PL T: Total (positions in m. from R1)

Maximum B.M. = 22.54 kNm at 0.50 m. from R1

Maximum S.F. = 45.19 kN at 0.00 m. from R1

Total deflection = $1.88 \times 10^8 / EI$ at 0.50 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 120 x 120 x 12.5 HF SHS S355 (40.9kg/m) $I_x = 982 \text{ cm}^4$ $r_y = 4.34 \text{ cm}$ $Z_x = 164 \text{ cm}^3$ $A = 52.1 \text{ cm}^2$ **Bending** $L_E/r_y = 1.00 \times 100/4.34 = 23$ $D/T = 9.6$ Permissible bending stress, $p_{bc} = 230.0 \text{ N/mm}^2$ (Table 2)Actual bending stress, $f_{bc} = 22.54 \times 1000/164 = 137 \text{ N/mm}^2$ OK**Shear**Section shear area = $0.5 \times CSA = 52.1 \times 100 \times 0.5 = 2,605 \text{ mm}^2$ (clause 23)Maximum shear stress, $f_s = 45.19 \times 1000/2,605 = 17.3 \text{ N/mm}^2$ OK

Web buckling and crushing have not been checked

DeflectionTotal deflection = $1.88 \times 1e8 / (205,000 \times 982) = 0.9 \text{ mm}$ ($L/1071$) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.372$ at 0.50 m. (≤ 1.25 OK) [14.c]

Beam: BC**Span: 6.5 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.81	0.81
U T	Floor	2*.4	0		L	2.60	2.60
U T	Partition	.45*3.2	0		L	4.68	4.68
						8.09	8.09

Total load: 16.19 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 13.15 kNm at 3.25 m. from R1

Maximum S.F. = 8.09 kN at 0.00 m. from R1

Total deflection = $57.9 \times 10^8 / EI$ at 3.25 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 203 x 102 x 23 UB S355D=203.2 mm B=101.8 mm t=5.4 mm T=9.3 mm $I_x=2,110 \text{ cm}^4$ $r_y=2.36 \text{ cm}$ $Z_x=207 \text{ cm}^3$ **Bending** $L_E/r_y = 6.50 \times 100/2.36 = 275$ D/T = 21.8Permissible bending stress, $p_{bc} = 65.2 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 13.15 \times 1000/207 = 63.5 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 8.09 \times 1000/(5.4 \times 203.2) = 7.4 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacity with load of 8.093 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 41.1 \text{ kN}$; $C2 = 1.40 \text{ kN/mm}$ Buckling: $p_c = 178 \text{ N/mm}^2$ (Table 17b); $C1 = 97.8 \text{ kN}$; $C2 = 0.963 \text{ kN/mm}$ Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 41.1 \text{ kN} <<<$ Buckling capacity, $P_x = C1 + L_b.C2 = 97.8 \text{ kN}$ **Deflection**Total deflection = $57.9 \times 1e8/(205,000 \times 2,110) = 13.4 \text{ mm}$ (L/486) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.949$ at 3.25 m. (≤ 1.25 OK) [14.c]**Bearings** (bearing plate sizing to BS5950-1:2000)203 x 102 x 23 UB stiff bearing length, $b_1 = t + 1.6r + 2T = 36.2 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: User-enteredFactored reaction = $8.09 \times 1.55 = 12.54 \text{ kN}$

Web of upper spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.

R2: 200 x 100 mm bearing plate

Factored reaction = $8.09 \times 1.55 = 12.54 \text{ kN}$

10 mm m.s. bearing plate, size 200 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)

Bearing plate projection beyond stiff bearing length = $(200 - 36.2)/2 = 81.9 \text{ mm}$

Factored stress under plate = $1.55 \times 8.09 \times 1000 / (200 \times 100) = 0.63 \text{ N/mm}^2$

Required plate thickness = $\sqrt{(3 \times 0.63 \times 81.9 \times 81.9 / 275)} = 6.78 \text{ mm}$: use 10mm

Factored bending stress in plate = $0.63 \times 81.9 \times (81.9/2) / (10 \times 10^6) = 126.3 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)

Beam: BD**Span: 4.1 m.**

	Load name	Loading w1	Start x1	Loading w2	End x2	R1comp	R2comp
U T	o.w.	0.25	0		L	0.51	0.51
U T	Floor	2*.4	0		L	1.64	1.64
U T	Partition	.45*3.2	0		L	2.95	2.95
						5.10	5.10

Total load: 10.21 kN

Load types: U: UDL T: Total (positions in m. from R1)

Maximum B.M. = 5.23 kNm at 2.05 m. from R1

Maximum S.F. = 5.10 kN at 0.00 m. from R1

Total deflection = $9.16 \times 10^{-8}/EI$ at 2.05 m. from R1 (E in N/mm^2 , I in cm^4)

Steel beam calculation to BS449 Part 2 using S355 steel

SECTION SIZE : 127 x 76 x 13 UB S355D=127.0 mm B=76.0 mm t=4.0 mm T=7.6 mm $I_x=473 \text{ cm}^4$ $r_y=1.84 \text{ cm}$ $Z_x=75.0 \text{ cm}^3$ **Bending** $L_E/r_y = 4.10 \times 100/1.84 = 223$ D/T = 16.7Permissible bending stress, $p_{bc} = 95.2 \text{ N/mm}^2$ (Table 3b)Actual bending stress, $f_{bc} = 5.23 \times 1000/75.0 = 69.8 \text{ N/mm}^2$ OK**Shear**Maximum shear in web, $f_s = 5.10 \times 1000/(4.0 \times 127.0) = 10.0 \text{ N/mm}^2$ OK**Beam web**

Check unstiffened web capacity with load of 5.104 kN

Bearing: $p_b = 260 \text{ N/mm}^2$ (Table 9); $C1 = 27.4 \text{ kN}$; $C2 = 1.04 \text{ kN/mm}$ Buckling: $p_c = 192 \text{ N/mm}^2$ (Table 17b); $C1 = 48.8 \text{ kN}$; $C2 = 0.769 \text{ kN/mm}$ Minimum required stiff bearing length, $L_b = 0 \text{ mm}$ Bearing capacity, $P_w = C1 + L_b.C2 = 27.4 \text{ kN} < < <$ Buckling capacity, $P_x = C1 + L_b.C2 = 48.8 \text{ kN}$ **Deflection**Total deflection = $9.16 \times 10^{-8}/(205,000 \times 473) = 9.4 \text{ mm}$ (L/434) OK**Combined bending and shear check (14.c)**Check $(f_{bc}/p_{bc})^2 + (f_s/p_s)^2 = 0.537$ at 2.05 m. (≤ 1.25 OK) [14.c]**Bearings (bearing plate sizing to BS5950-1:2000)**127 x 76 x 13 UB stiff bearing length, $b_1 = t + 1.6r + 2T = 31.4 \text{ mm}$

Factor reactions by 1.55 (user selected value)

R1: 125 x 100 mm bearing plateFactored reaction = $5.10 \times 1.55 = 7.91 \text{ kN}$

10 mm m.s. bearing plate, size 125 x 100 mm

Local design strength of masonry (factored) = 0.700 N/mm^2 (User-entered value)Bearing plate projection beyond stiff bearing length = $(125-31.4)/2 = 46.8 \text{ mm}$ Factored stress under plate = $1.55 \times 5.10 \times 1000/125 \times 100 = 0.63 \text{ N/mm}^2$ Required plate thickness = $\sqrt{(3 \times 0.63 \times 46.8 \times 46.8/275)} = 3.89 \text{ mm}$: use 10mmFactored bending stress in plate = $0.63 \times 46.8 \times (46.8/2)/(10 \times 10/6) = 41.6 \text{ N/mm}^2$ ($p_y = 275 \text{ N/mm}^2$)**R2: User-entered**

Factored reaction = $5.10 \times 1.55 = 7.91$ kN

Web of Upper spine wall beam via 2 No 100x100x10 thick rt angle brackets and 2 No M16 bolts per face.